LABORATORY REPORT

Application Development Lab
(CS33002)

B.Tech Program in CSE

Submitted By

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Experiment Number	3

Experiment Title	Pneumonia Detection using CNN
Date of Experiment	21/01/2025
Date of Submission	28/01/2025

1. Objective: Understand the basics of machine learning and Python/R

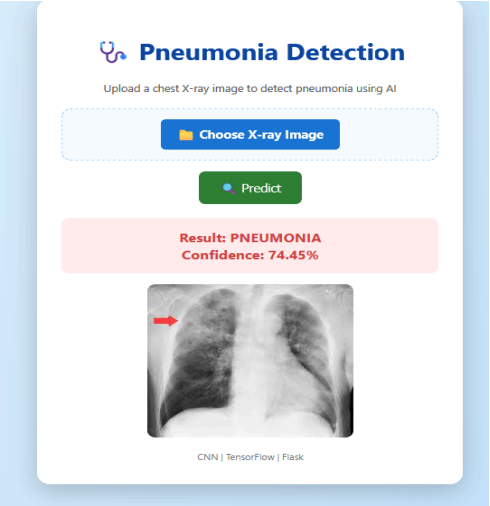
2. Code:-

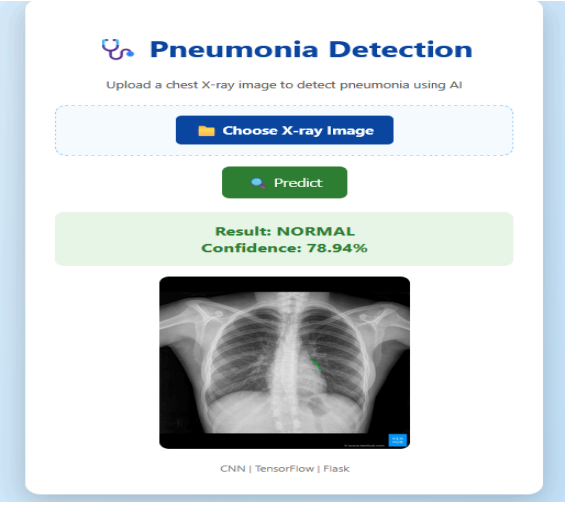
```
import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D, Dropout
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
from sklearn.utils.class_weight import compute_class_weight
import numpy as np
import matplotlib.pyplot as plt
IMG_SIZE = 224
BATCH_SIZE = 32
EPOCHS = 15
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    zoom_range=0.2,
    horizontal_flip=True)
val_test_datagen = ImageDataGenerator(rescale=1./255)
train_gen = train_datagen.flow_from_directory(
    "chest_xray/train",
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='binary')
val_gen = val_test_datagen.flow_from_directory(
    "chest_xray/val",
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='binary')
test_gen = val_test_datagen.flow_from_directory(
    "chest_xray/test",
```

```
target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='binary',
    shuffle=False)
class_weights = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(train_gen.classes),
    y=train_gen.classes)
class_weights = dict(enumerate(class_weights))
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(IMG_SIZE, IMG_SIZE, 3))
base_model.trainable = False  # IMPORTANT
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.5)(x)
output = Dense(1, activation='sigmoid')(x)
model = Model(inputs=base_model.input, outputs=output)
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),
    loss='binary_crossentropy',
    metrics=['accuracy'])
model.summary()
early_stop = EarlyStopping(patience=3, restore_best_weights=True)
checkpoint = ModelCheckpoint(
    "pneumonia_mobilenet.keras",
    monitor='val_accuracy',
    save_best_only=True)
history = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=EPOCHS,
    class_weight=class_weights,
    callbacks=[early_stop, checkpoint])
loss, acc = model.evaluate(test_gen)
print(f"\n Test Accuracy: {acc*100:.2f}%")
plt.figure(figsize=(12,5))plt.subplot(1,2,1)
plt.plot(history.history['accuracy'], label='Train')
```

```
plt.plot(history.history['val_accuracy'], label='Validation')
plt.legend()
plt.title("Accuracy")
plt.subplot(1,2,2)
plt.plot(history.history['loss'], label='Train')
plt.plot(history.history['val_loss'], label='Validation')
plt.legend()
plt.title("Loss")
plt.show()
```

3. Results/Output:- Entire Screen Shot including Date & Time





4. Remarks:-

Signature of the Student _____ (Name of the Student)	Signature of the Lab Coordinator _____ __ (Name of the Coordinator)